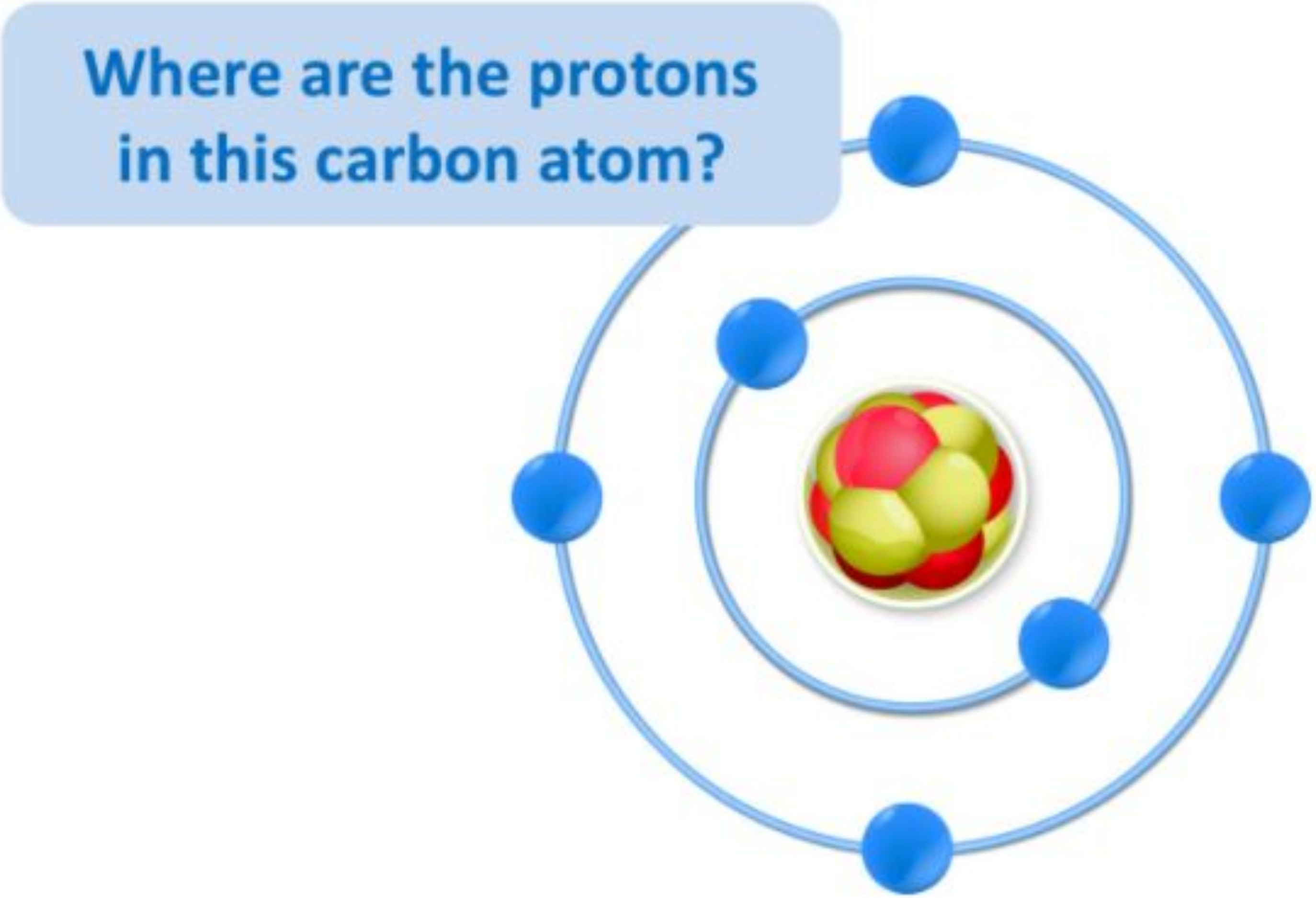
Title – Chemical Measurements and Uncertainty



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DO NOW

Chemical Measurements and Uncertainty

Lesson 4

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HOME LEARNING

Learning objectives:

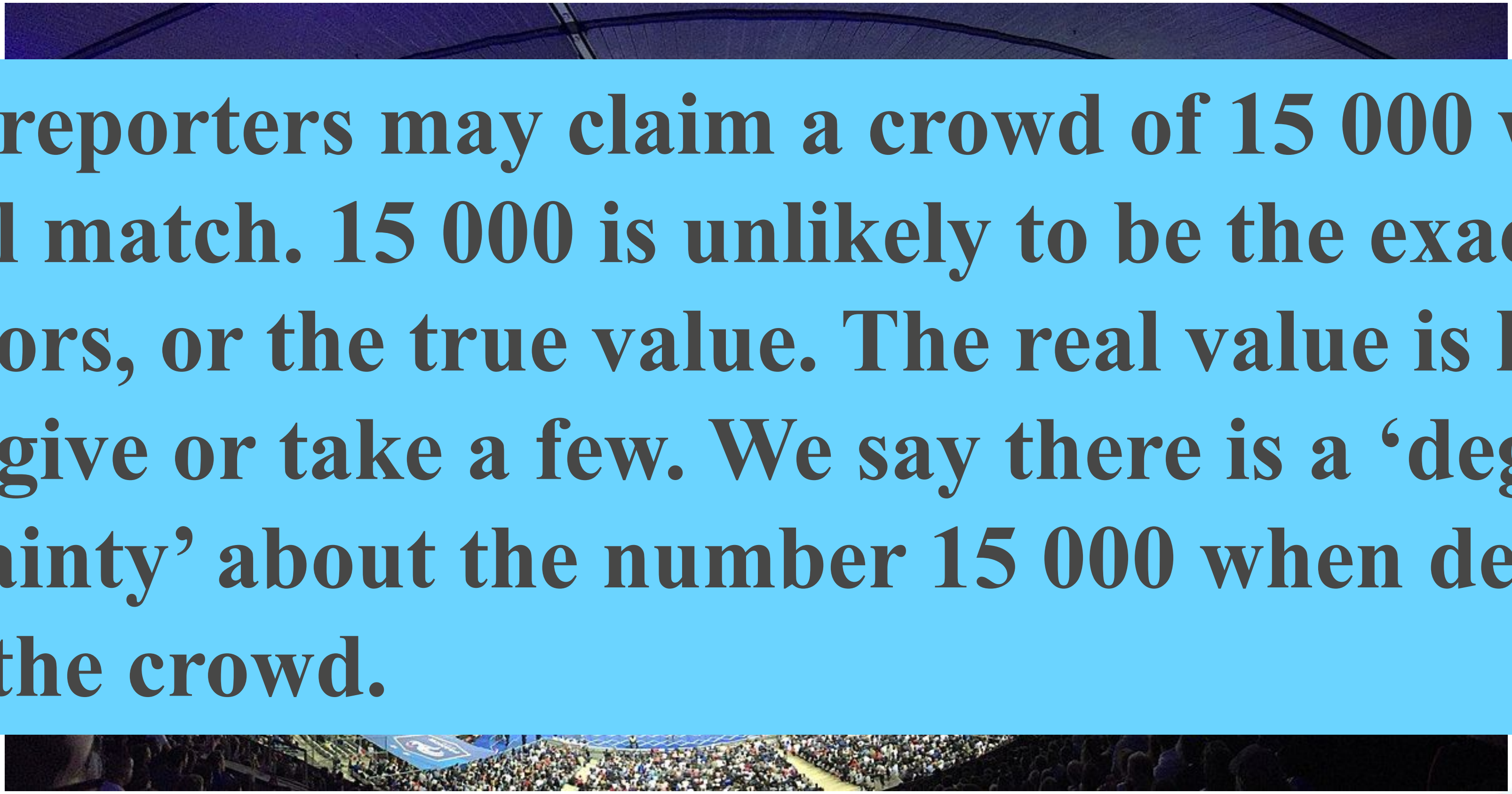
By the end of the lesson you will be able to:

- understand that all measurements have a degree of uncertainty
- estimate the uncertainty from the distribution of results
- measure uncertainty from the range of a set of measurements and their mean.

Key Words

uncertainty
distribution of results
range of measurements

How many people are in this stadium?



Sports reporters may claim a crowd of 15 000 watched a football match. 15 000 is unlikely to be the exact number of spectators, or the true value. The real value is likely to be 15 000 give or take a few. We say there is a 'degree of uncertainty' about the number 15 000 when describing the size of the crowd.

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Chemical measurements

- Whenever a measurement is made in chemistry, there is always some uncertainty in the result obtained. Are the results valid or reliable?

- There are many causes of uncertainty in chemical measurements. For example it may be difficult to judge: What causes these invalid results?

1. Whether a thermometer is showing a temperature of 24.0°C, 24.5°C or 25.0°C

2. Exactly when a chemical reaction has finished

3. Whether factors were completely controlled to not have an impact on the results

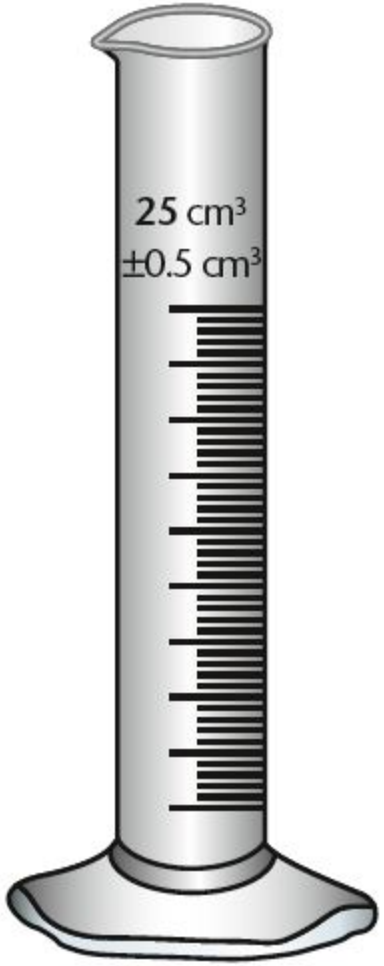
- There are two ways of estimating uncertainty:

1. **By considering the resolution of measuring instruments**
2. **From the range of a set of repeat measurements**

What can we use to measure a volume of liquid?

Measuring cylinders come in different sizes. A 50 cm^3 cylinder will have a scale with 1 cm^3 divisions. You can probably read volumes to half a division, or to 0.5 cm^3. There is always an **uncertainty** when a measurement is made. This is usually printed on the apparatus.

A 25 cm^3 measuring cylinder has an uncertainty of ±0.5 cm^3. A volume of 15.0 cm^3 should really be written as 15.0 cm^3 ± 0.5 cm^3. This means that the true measurement will be somewhere between 14.5 cm^3 and 15.5 cm^3.



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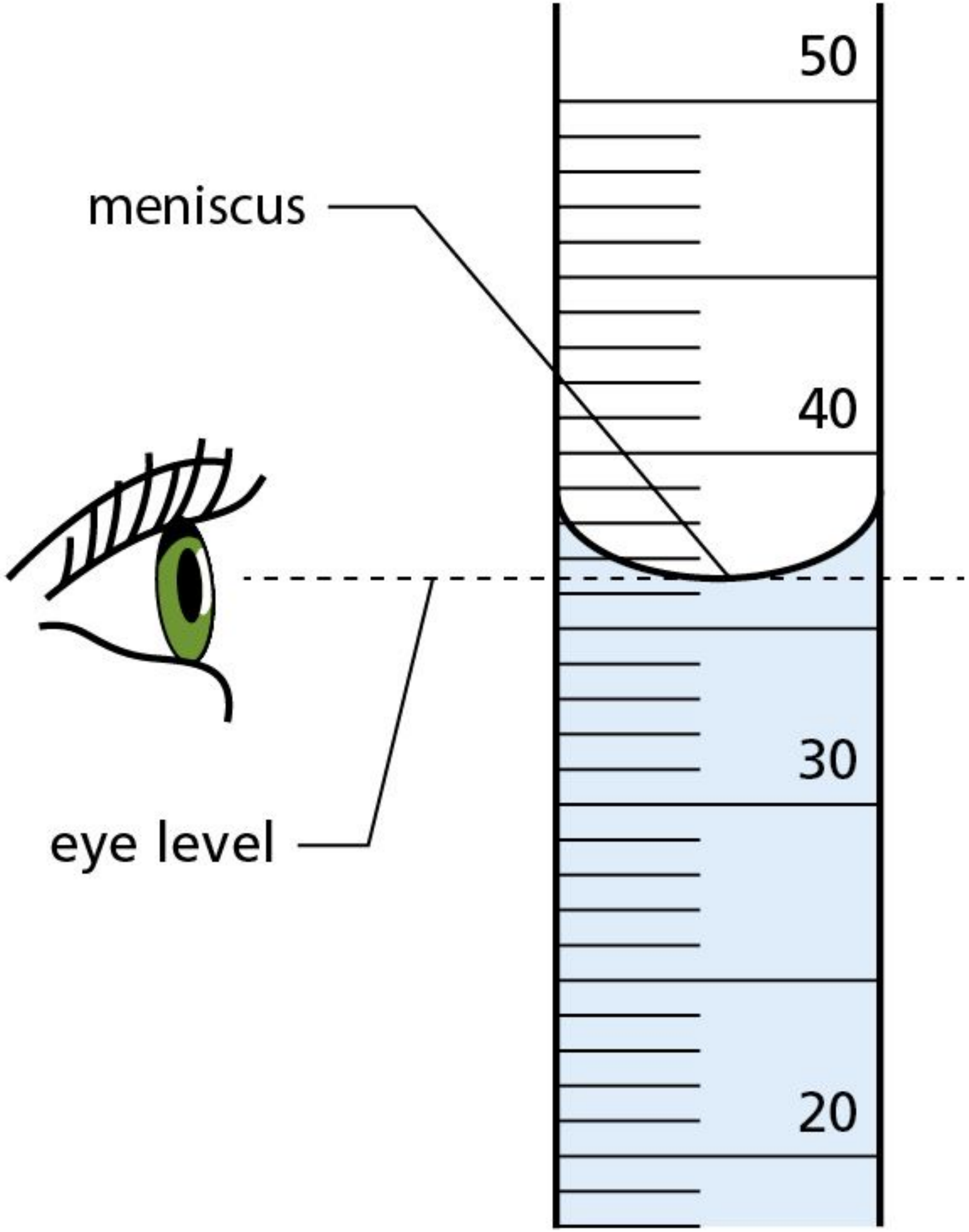
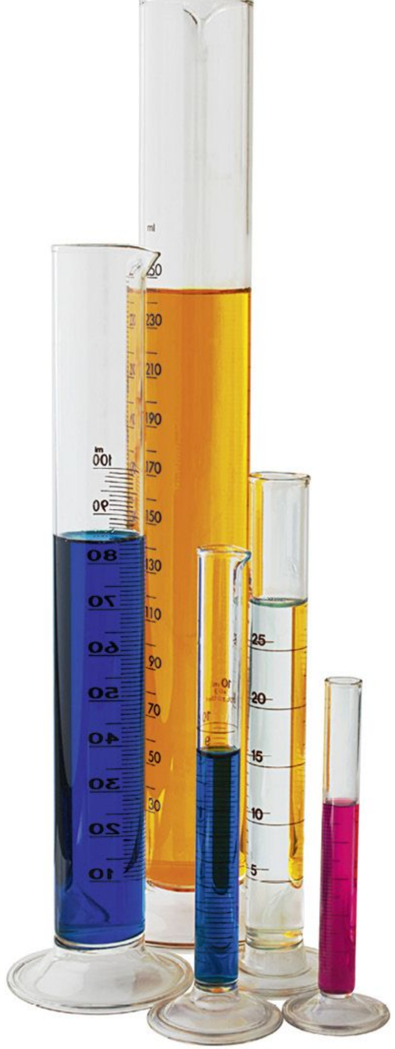
A burette used to measure volumes of liquids has an uncertainty of ±0.05 cm^3. Emma uses it to measure 25.30 cm^3 of water. How much water could Emma actually have measured?

Between 25.25 and 25.35 cm^3

Liquids are not as easy to measure as they may seem because they do not form a straight line at the surface. The surface curves downwards – this is called a **meniscus**. Scientists use the bottom of the meniscus to make their readings. They do this at eye level so that they can be sure that they are reading the position and scale accurately.





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The **range** of a set of measurements is the difference between the highest and lowest measurements.

range = highest measurement − lowest measurement

These are some class results for a titration experiment:

6.8 g/dm^3 | 7.1 g/dm^3 | 6.9 g/dm^3 | 8.2 g/dm^3 | 6.3 g/dm^3 | 7.4 g/dm^3 | 6.6 g/dm^3 | 5.9 g/dm^3

- the range of the results = 8.2 – 5.9 = 2.3 g/dm^3

The mean result is found by adding all the results together and dividing by the number of results:

- mean result = 6.9 g/dm^3

The range and mean can be used to calculate the percentage uncertainty.

$$\text{percentage uncertainty} = \frac{\text{range of measurements}}{\text{mean}} \times 100$$

- the percentage uncertainty for the class results is 33.3%

Have a practice

Test number	1	2	3	4	5
Volume	24.0	24.5	23.5	25.0	23.0

Answer

mean = $\frac{24.0 + 24.5 + 23.5 + 25.0 + 23.0}{5}$

= 24.0 cm^3

range = (biggest value - smallest value)

= 25.0 - 23.0

= 2.0 cm^3

$$\boxed{uncertainty = \frac{range\ of\ results}{2}}$$

=

= ± 1.0 cm^3

So the volume is 24.0 cm^3 ± 1.0 cm^3.

Calculating Uncertainty

$$uncertainty = \frac{range\ of\ results}{2}$$

The **range** is the difference between the **largest** and **smallest** value from a set of repeated tests.

For example, Ami carried out a practical to determine how much dilute hydrochloric acid is needed to react with exactly 30.0cm^3 of a sodium hydroxide solution.

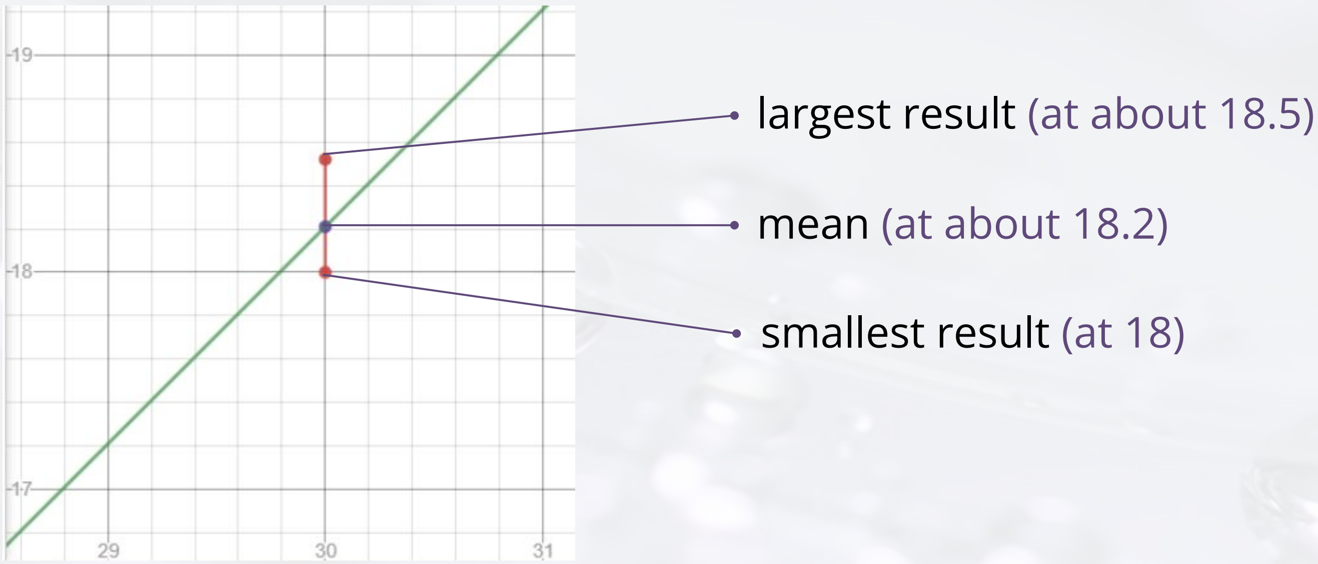
Repeat	1	2	3	Mean
Volume of HCl needed to react with 30.0cm^3 of NaOH.	18.12	18.52	18.00	

1. Calculate the mean of the results. **(18.12 + 18.52 + 18.00) ÷ 3 = 18.21**
2. Calculate the range. **18.52 – 18.00 = 0.52cm^3**
3. Calculate the uncertainty of the mean. **0.52 ÷ 2 = 0.26cm^3**
4. The mean with uncertainty is written as **18.21 ±0.26cm^3**

Calculating Uncertainty

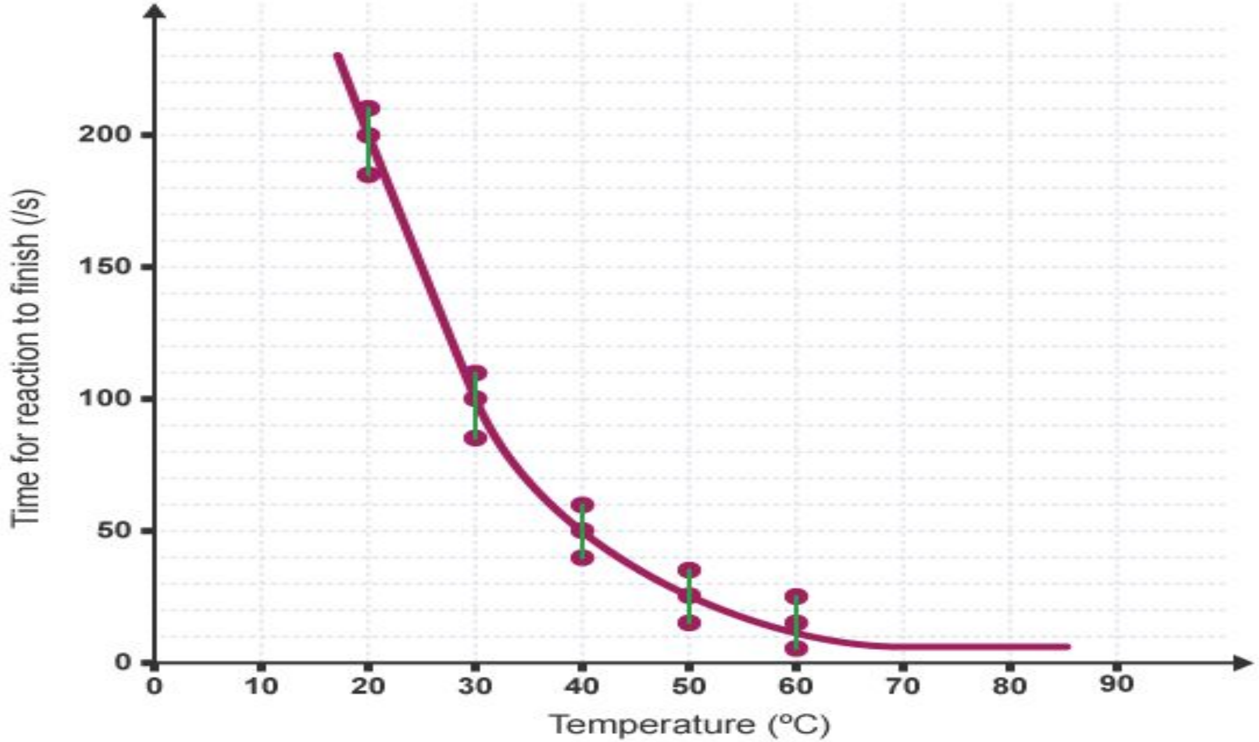
Uncertainty can also be displayed on a graph. The mean is taken to represent the set of repeated results but the smallest and largest values are also shown.

Repeat	1	2	3	Mean
Volume of HCl needed to react with 30.0cm^3 of NaOH.	18.12	18.52	18.00	18.21



Showing uncertainty in a graph

- Uncertainty can also be shown on a graph. All the repeat readings for each value of the independent variable are plotted.
- Vertical lines joining these values represent the uncertainty.



Errors that occur in experiments

- There are 2 types of errors made during an experiment:
- 1. Random error- Random errors may be caused by human error such as a poor technique when taking measurements or by equipment that is faulty
- 2. systematic error- errors that are produced consistently. If the experiment is repeated, the same error will occur.

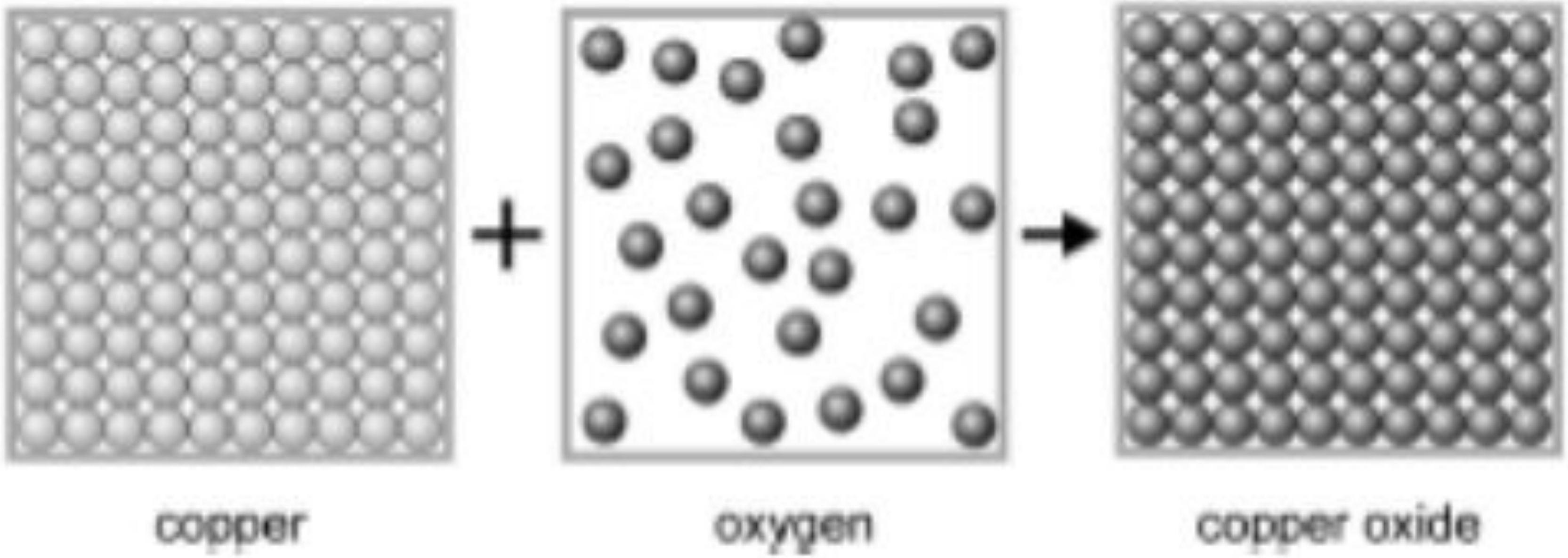
Plenary

3. What is the difference in meaning between 'range' and 'mean'?

4. A set of measurements has a range of 0.60 cm and a mean of 7.5 cm. What is the percentage uncertainty?

1. Draw three boxes. Draw the particle arrangement for the reactants and products in the reaction of:

copper + oxygen → copper oxide



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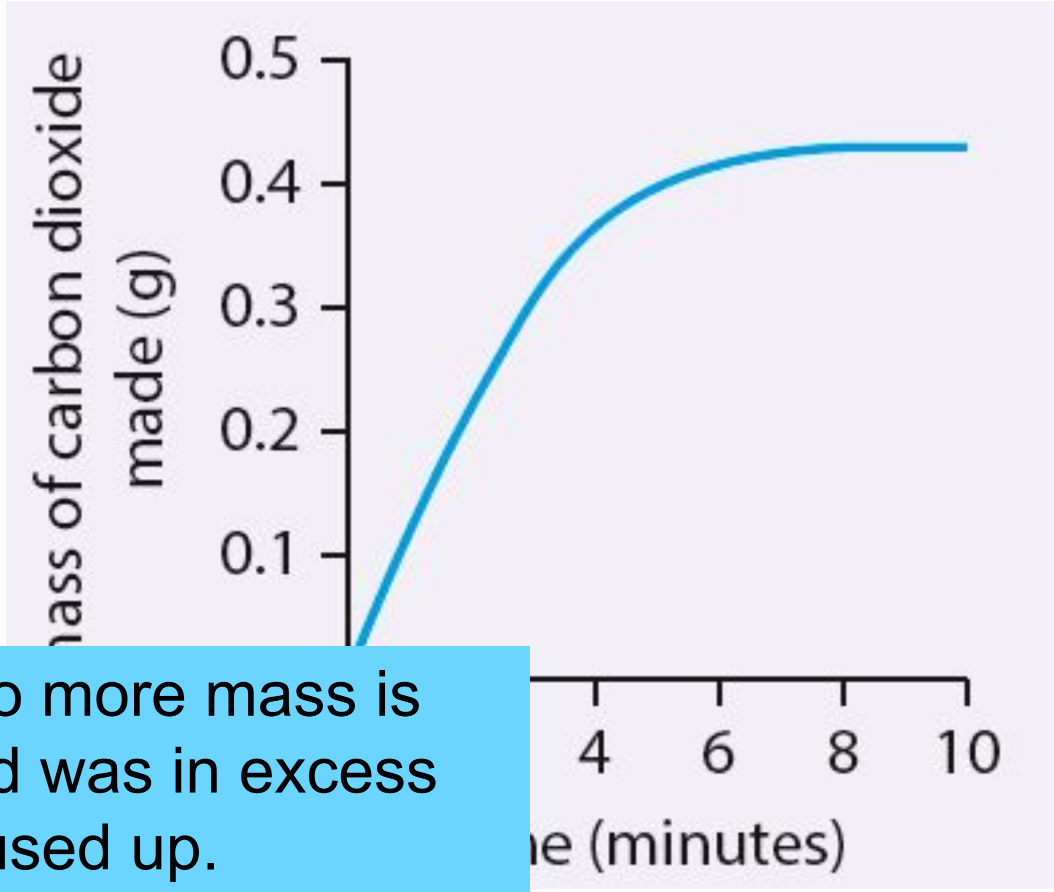
1. Look at the graph. Explain when the reaction stopped and why it stopped.
2. Explain whether mass will be gained or lost in the following reaction. CH_3COOH and H_2O are liquids and O_2 and CO_2 are gases $CH_3COOH + 2O_2 \rightarrow 2CO_2 +$



1. 8 minutes. Graph Is horizontal meaning no more mass is being lost so the reaction is finished. The acid was in excess and all the magnesium carbonate has been used up.

2. Mass of O_2 gained = 2 x 32 = 64

Mass of CO_2 lost = 2 x 44 = 88

So 88 – 64 = 24g of mass lost overall

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